Presentation On

Earthquake Resistive Loading Bearing Structure

Presented by:

Er. Ramesh Thing

MOUD, DLIPU, Ramechhap

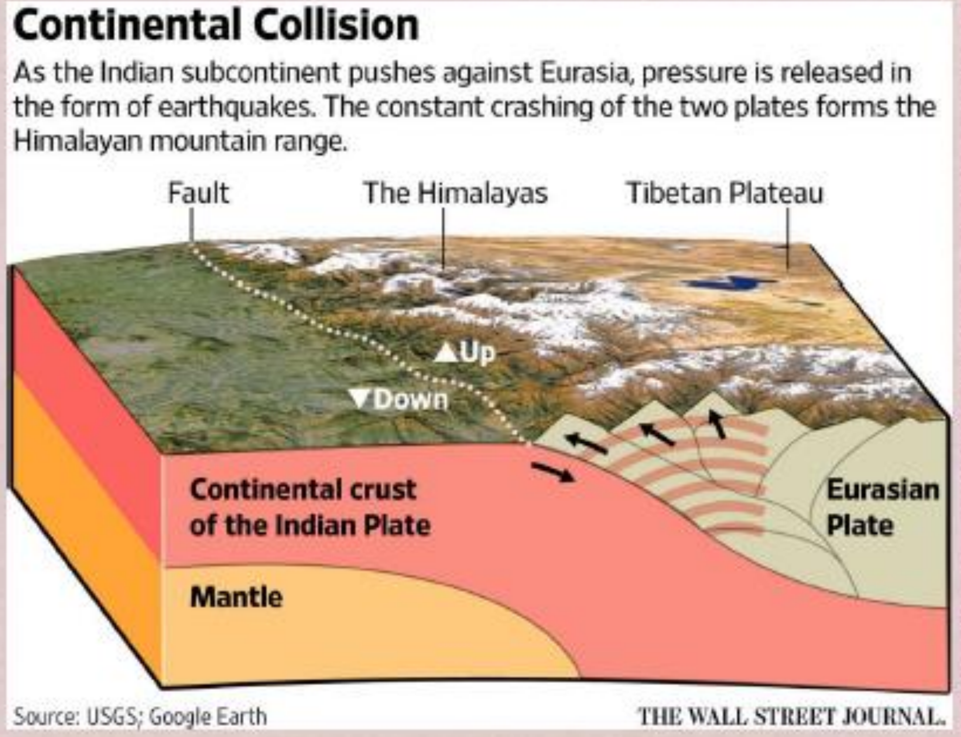
EARTHQUAKE…..!!!!!! 😱

भोइ +चल्यो = भुईचालो

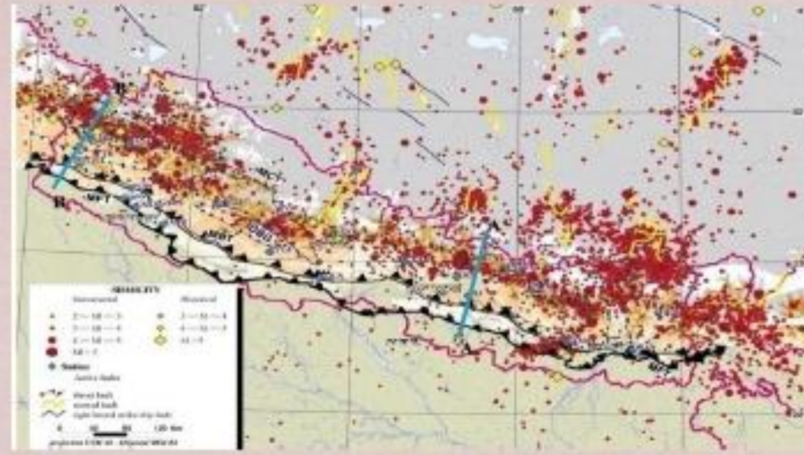
According to Wikipedia **"an earthquake is the perceptible shaking of the surface of the Earth, resulting from the sudden release of energy in the Earth's crust that creates seismic waves."**

How Earthquake Happen?

- Movement Tectonic plate
- Volcanic eruption
- Nuclear activities

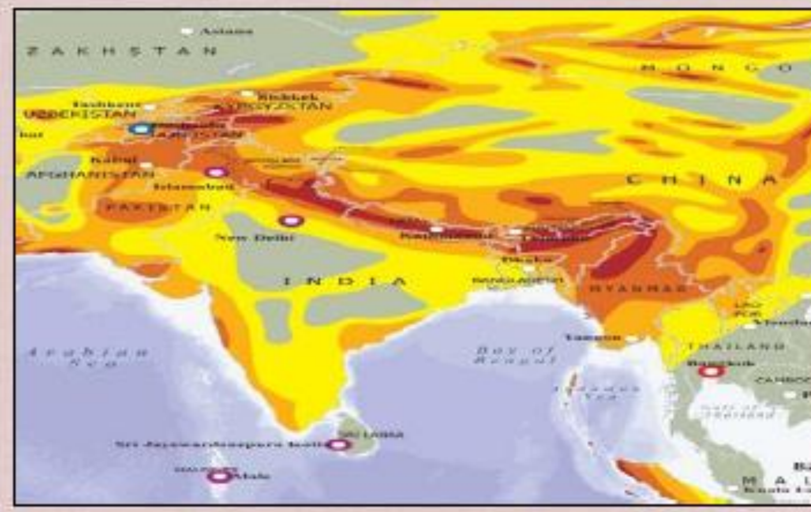


Seismicity of Earthquake



- Nepal is located at the boundary between two large plates, Indian plate at the south and Tibetan or Eurasian plate at the north. Indian Plate is sub ducting underneath the Tibetan plate at an average of 25 mm per year (ADPC and NSET, 2005).

- Our mountains are the products of these tectonic movements. Due to its location at the sub ducting zone between two large plates Nepal and the entire Himalayan region is highly earthquake prone



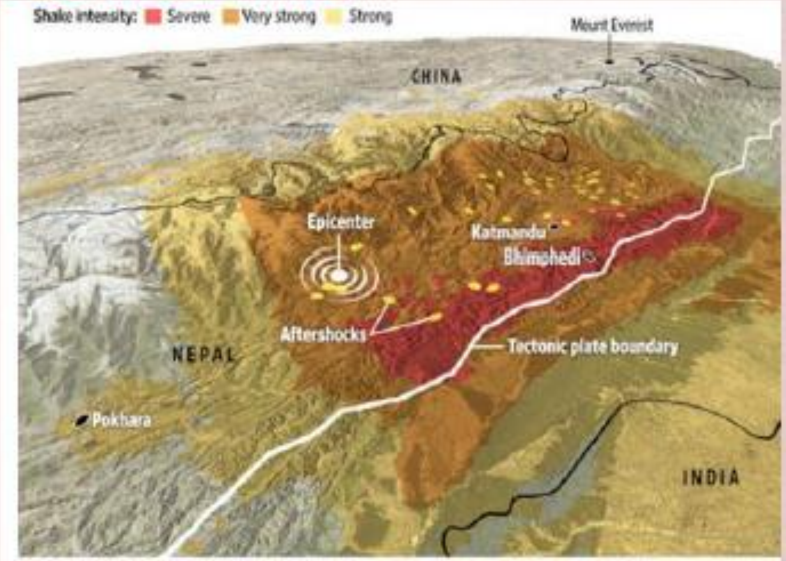
Seismic Hazard map of Asia also shows our whole Himalayan region as the highly seismic area

Earthquake 2072

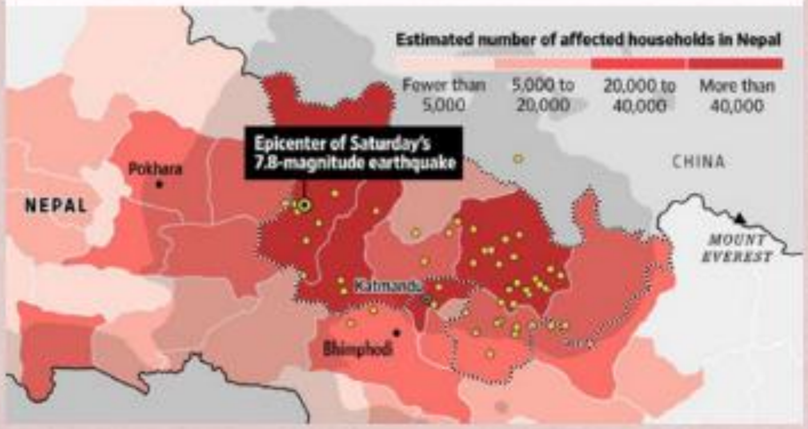
- Date :April 25, 2015 (Baishak 12, 2072)
- Magnitude : 7.8 earthquake
- epicenter: Gorkha
- Depth: 11 KM beneath earth surface.
- Major affected areas:

Gorkha
Kathmandu,Bhaktapur,Lalitpur,Kavre,
Dhading,Nuwakot, **Ramechhap**,
Sindhuli, Okhaldunga,
Rasuwa,Dolakha,Sindhupalchowk

- About 9000 people die
- 602,257 houses were fully damaged,
- 285,099 houses were partially damaged.



Seismic Landscape



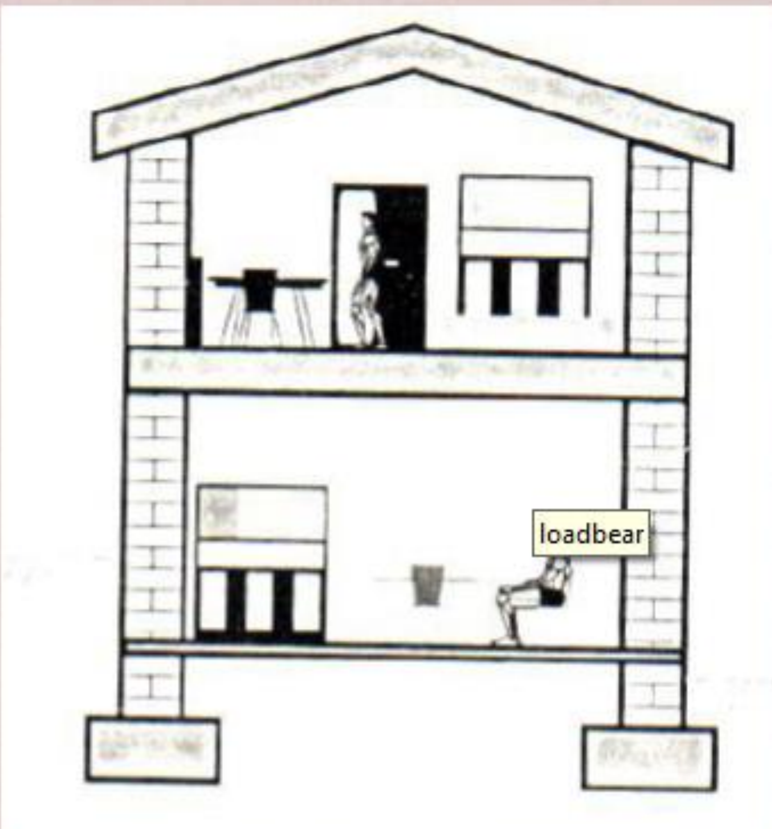
Estimated damaged household

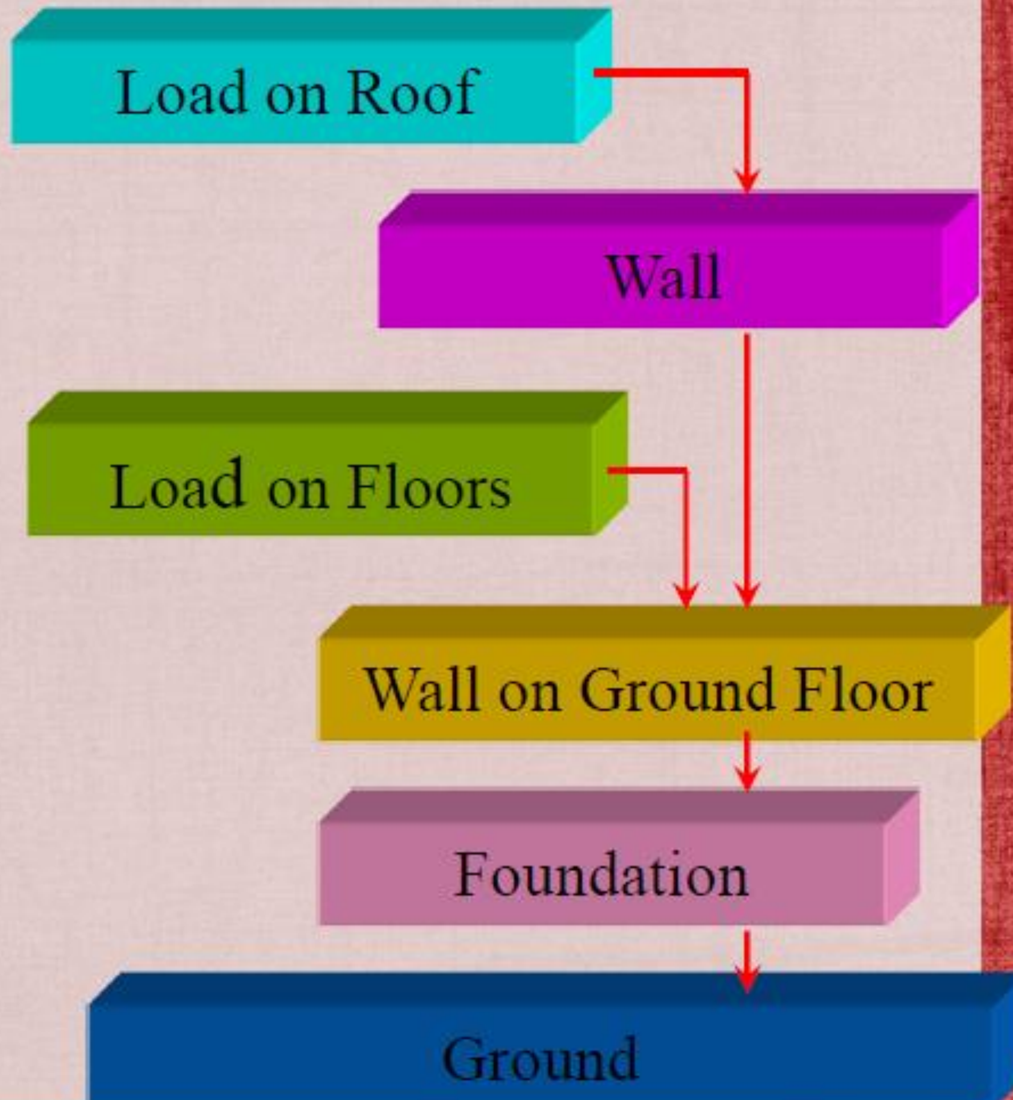
Loadbearing Structure

- A load-bearing wall or bearing wall is a wall that bears the weight of the house above said wall, resting upon it by conducting its weight to a foundation structure. The materials most often used to construct load-bearing walls in large buildings are concrete, block, or brick.

Vertical Load Path

LOAD BEARING SYSTEM





Building destruction and damage due to earthquake

Performance of LB Buildings

Complete collapse of LB buildings in Bhaktapur

Collapse of LB buildings at yambhu, Kathmandu

- Old unreinforced masonry buildings suffered extensive damage due to **low strength of masonry via material deterioration over the years**

Corner Separation Leading to Failure



Corner separation of URM buildi

- Loss of strength has been observed due to the formation of vertical cracks at the corners of the building, which reduced the **out-of-plane stability** of the walls significantly
- In many cases, damage occurred due **to absence of sill and lintel bands** around openings
- Formation of **shear cracks** has been observed over the entire storey height
- Flexible diaphragms deteriorated over the years causing collapse of floor as well as failure of supporting walls

Performance of LB (Stone) Buildings

Sindhupalchowk: Building with wooden ties survived the full collapse

- Out-of-plane failure
- Full collapse prevented due to the presence of horizontal and vertical wooden ties

Failure of Gable Walls

Failure of gable walls due to the high height and lack of support at the top


CITY OF GOD !!!!!!!!!
OR
CITY OF mud !!!!!!!!!

Earthquake and loadbearing structure

WHY ???????

Failure of Masonry Walls





- Out-of-plane collapse of load bearing masonry walls of old buildings was widely observed in many parts
- Poor connection with the floor diaphragm and cross walls led to such collapse

Performance of LB Buildings

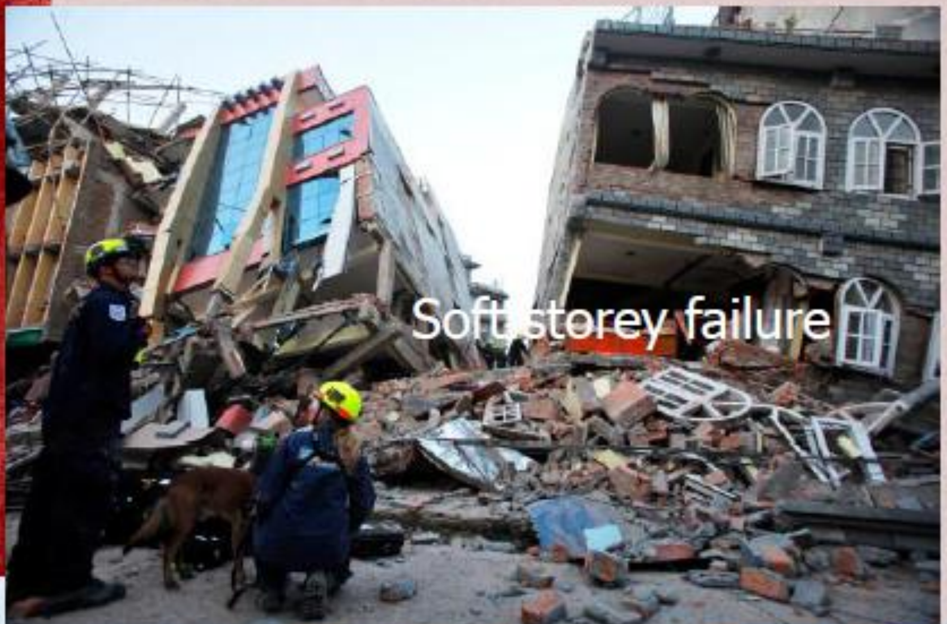
Sanga chowk-Sindhupalchowk: Building with wooden ties Fully collapsed and Building with PCC sill also fully collapsed

- Full collapse of stone masonry building, only timber posts are standing (loss of life)
- Single storey school bldg with plinth tie, RCC lintel band, vertical RCC posts at corners, steel truss and CGI sheet roofing had collapsed. Guess level of shaking?
- Buttress walls have been placed to increase out-of-plane bending

Vertical Cracks and Separation of Walls

Corners in masonry buildings are inherently week and first suffer from horizontal force during Earthquake.

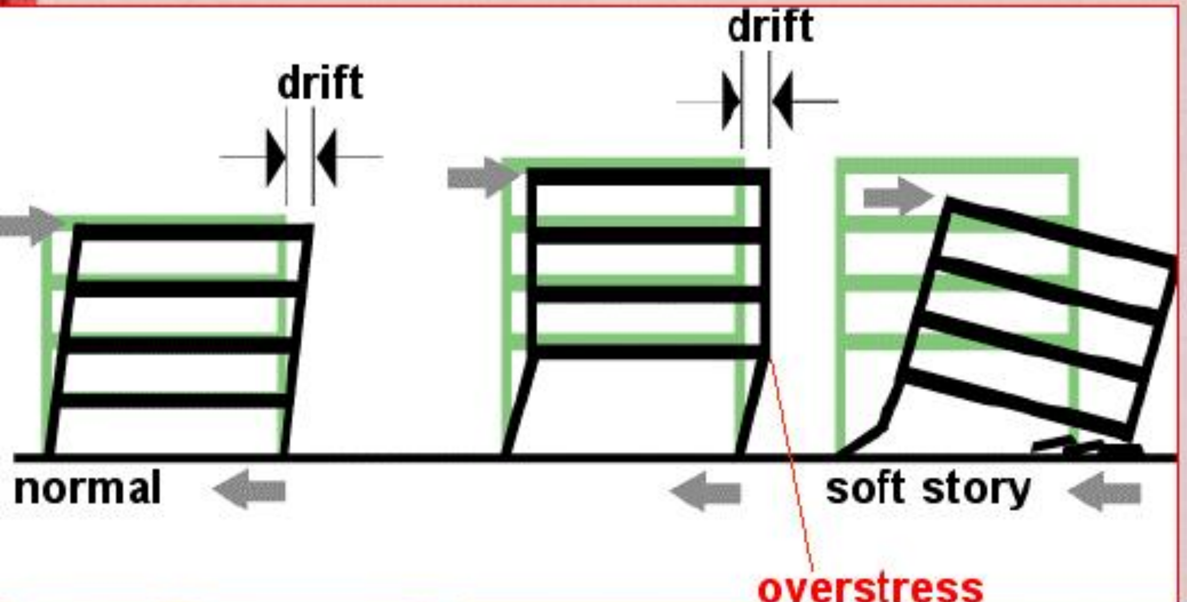
Performance of RC Buildings



Pounding with soft storey effect

- Inadequate column size
- Poor reinforcement detailing of columns, beams and beam-column joints
- Violation of design philosophy, "**Strong column-weak-beam principle**"

Soft Storey Effect



- ➢Stress concentration at the ground floor columns
- ➢Poor performance due to inadequate column size
- ➢Poor ductile detailing of reinforcement bars in columns / beam-column joints

Soft storey failure

Pounding/Soft Storey Failure

E:\2015 nepalEQ\earthquake 2072 pics from bivek\From Birendra Khadka\RCC Structures\Storey collapse\IMG_6478.JPG

ing effect compounded with soft storey failure

- Violence of design principle

" Strong-column-weak- beam"

- Poor reinforcement detailing

Pounding/Impact



- Row housing is common in urban areas: leading to pounding since the buildings are of different dynamic properties
- Impact of adjacent structure aggravated damage/collapse in most cases

Short Column Effect

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- Special confining shear reinforcement is to be provided over the whole height of column that are likely to sustain short column effect

Basin Edge Effects

- More damage along the northern part of basin: Naikap; Swichatar; Ramkot; Swyombhu
- Lack of strong ground motion stations along edge
- Level of shaking is high

Damage at Central Fisheries Building

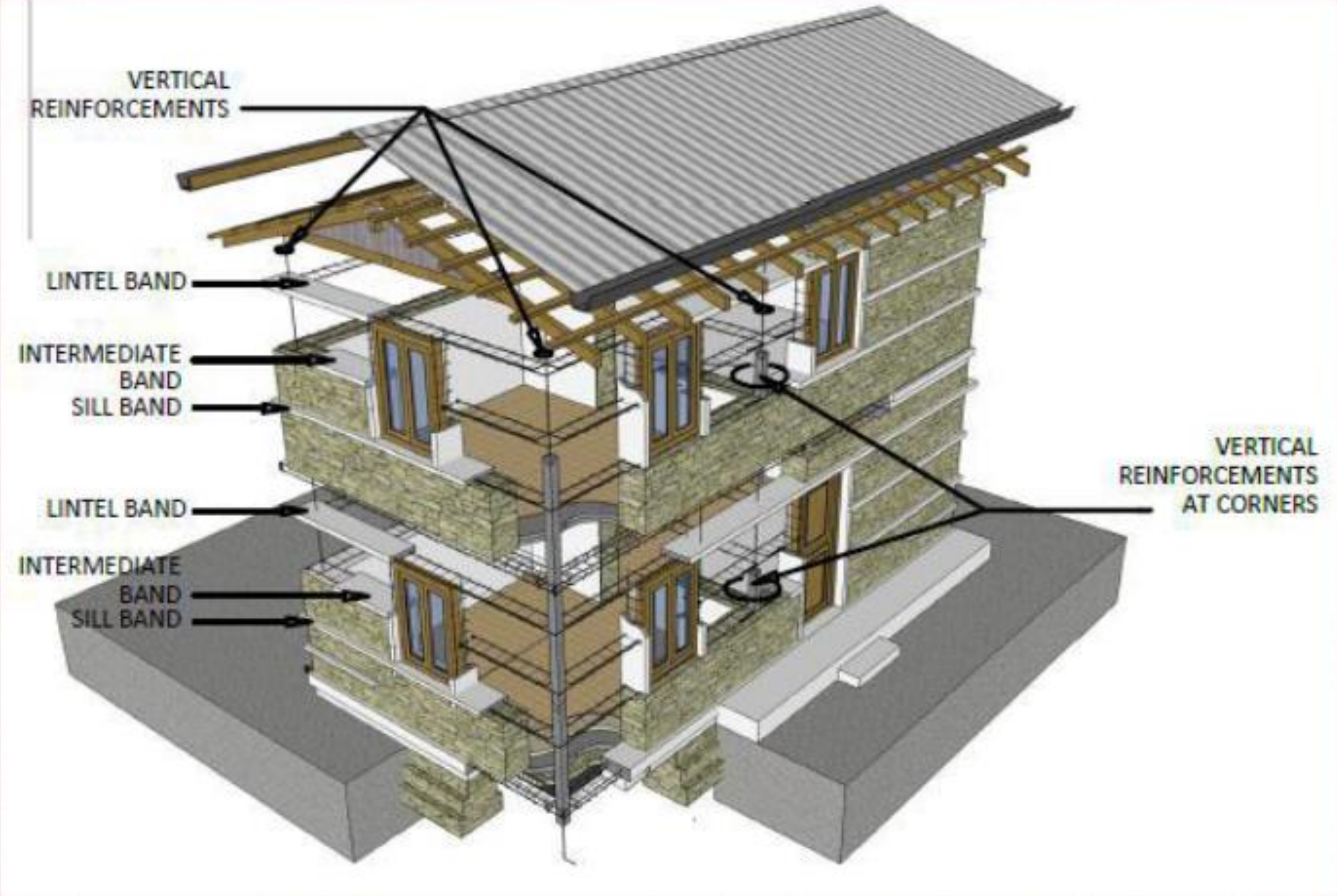
➤ Local site amplification: Estimation of GM via modeling of building

Failure of GF Columns (450mmx450mm)

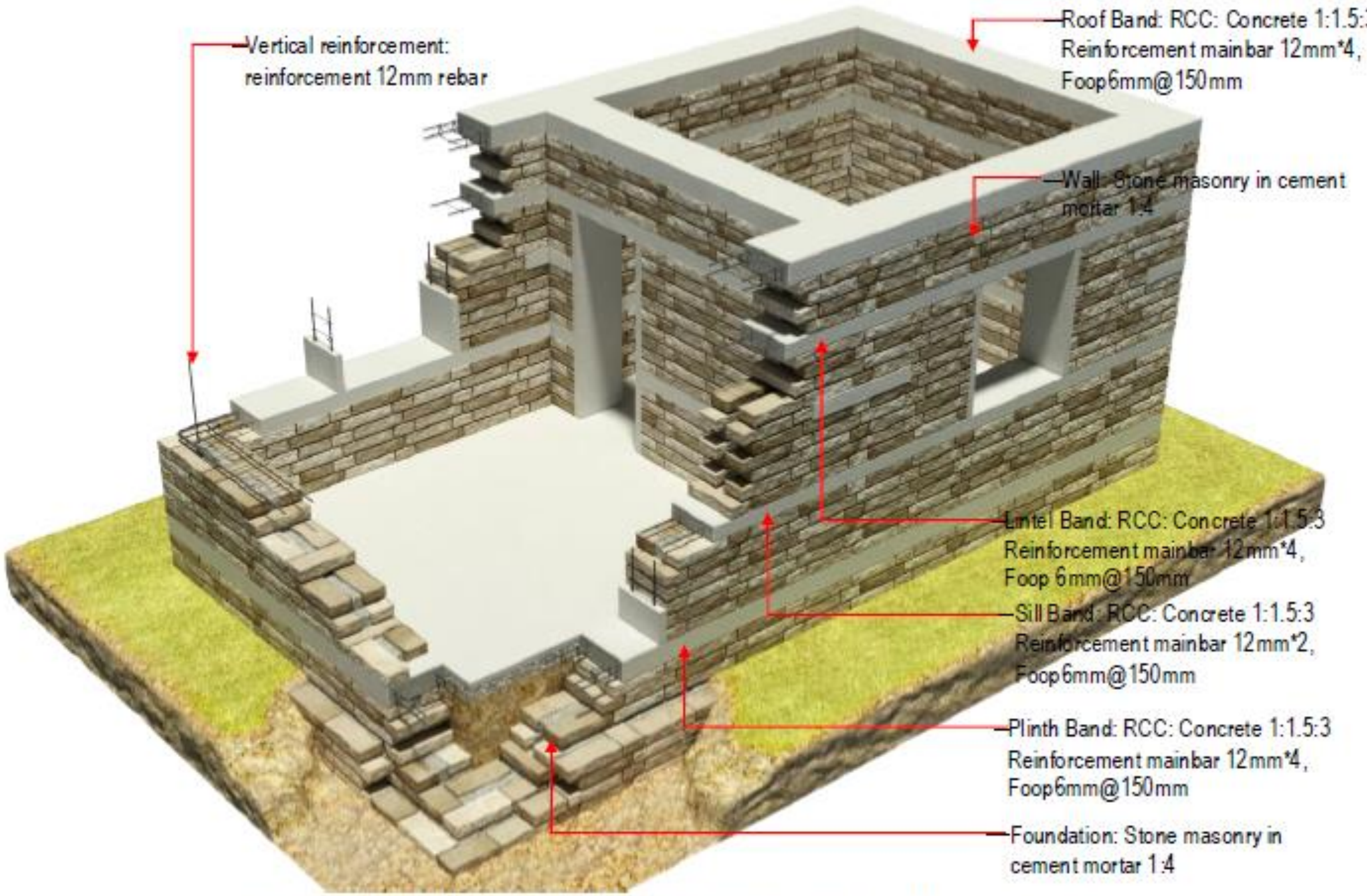
Repair/Rehabilitation works

- Damage assessment has been carried out
- Building capacity assessment?

Structural member of Earthquake Resistive loadbearing



Structural member(load bearing)



Technology for Earthquake Resistant Building Construction (Stone in Cement Mortar)

Improvement Of Loadbearing Structure ....

Location of Building: Earthquake Zone

Soil Conditions

Slope 0f Land

Shape 0f Building

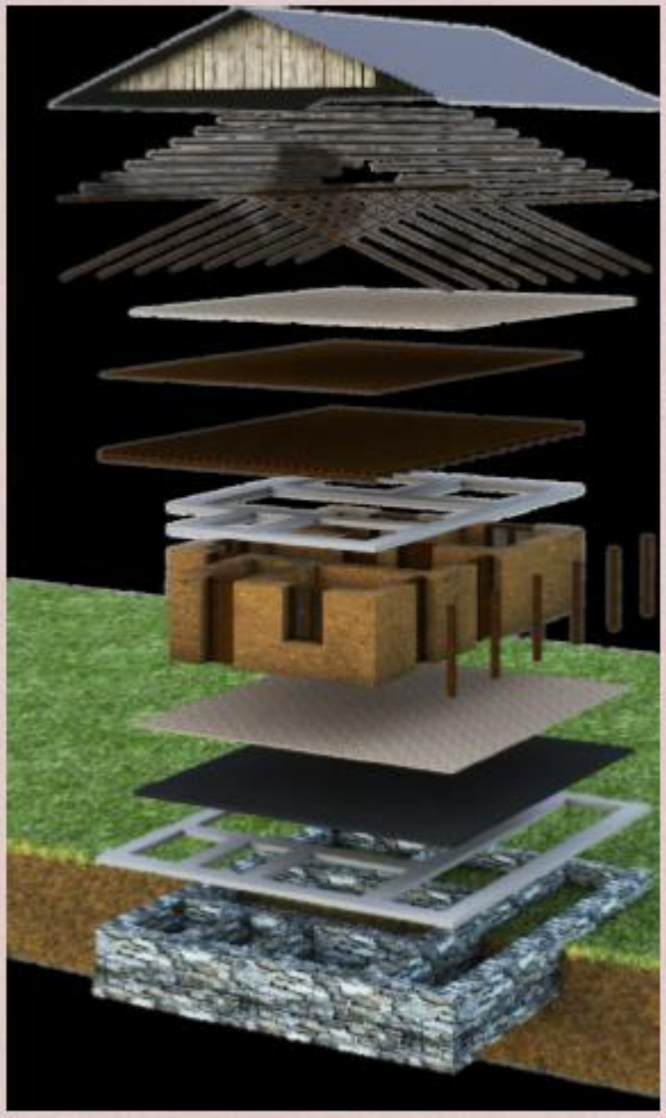
Horizontal Ratio

Vertical Ration

Building material

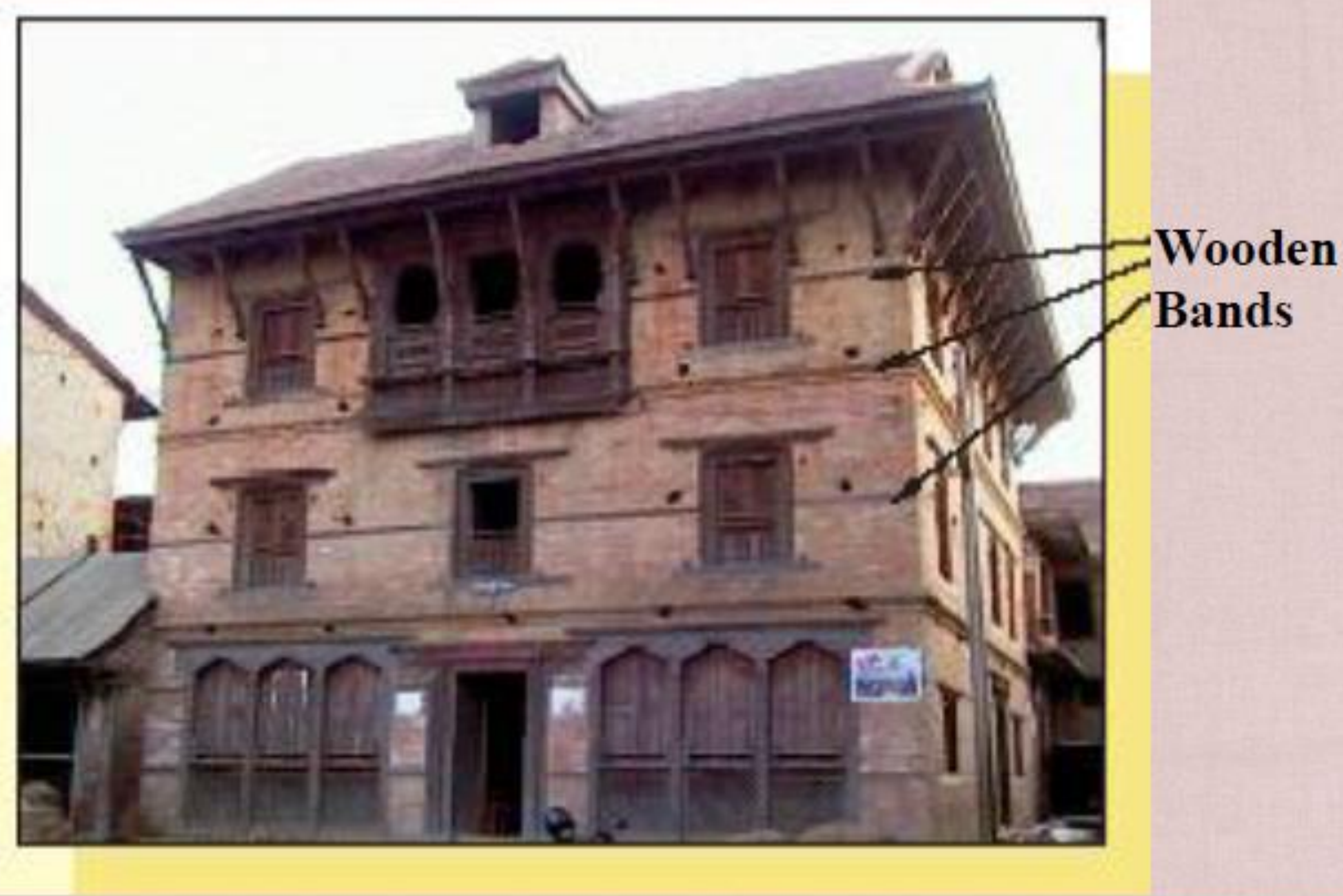
Construction technology

Skilled manpower



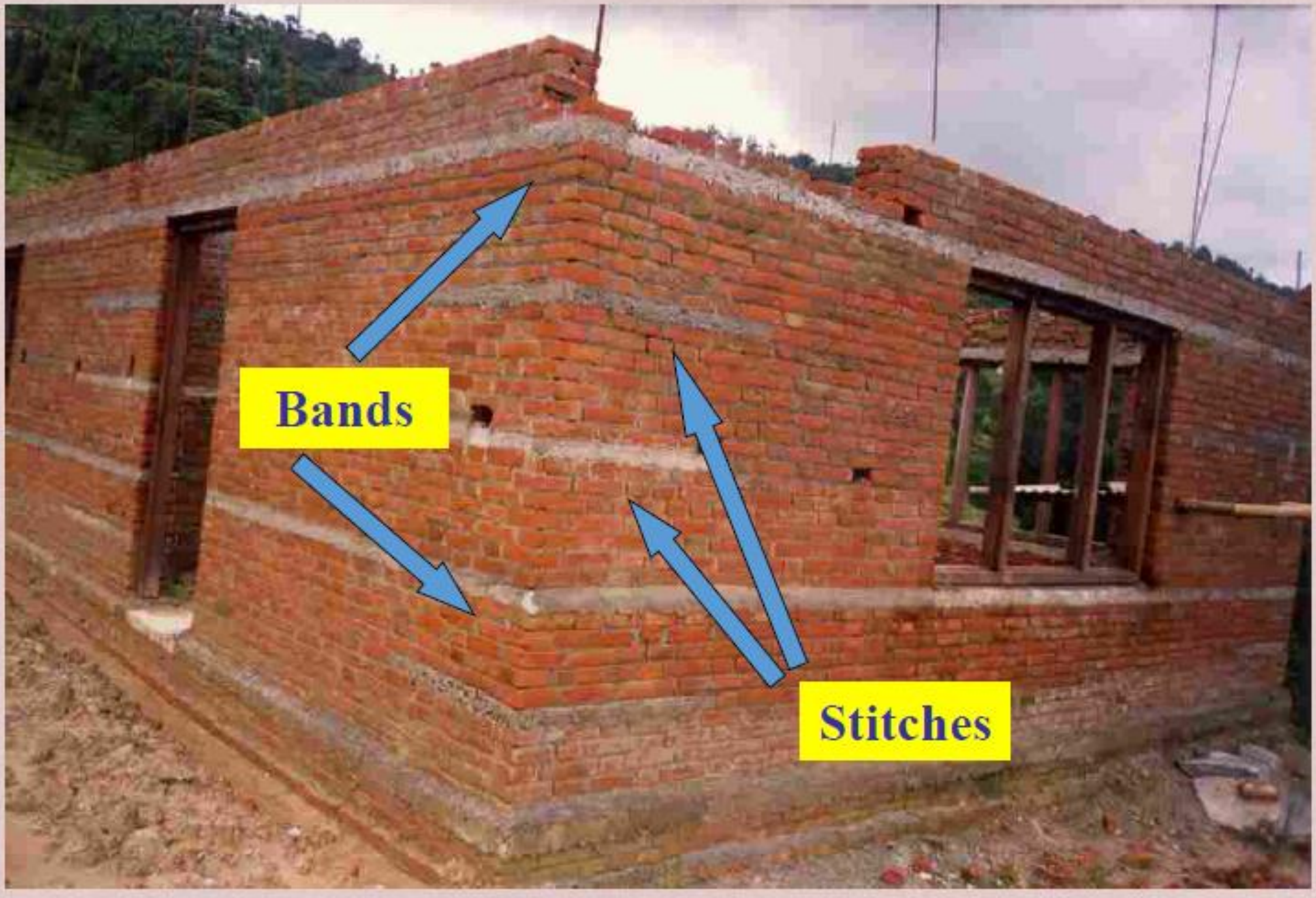
Improvement Of Loadbearing Structure …….

Earthquake Resistance Load Bearing building (using Wooden Bands)



Earthquake Resistance Load Bearing Building
(using concrete band)

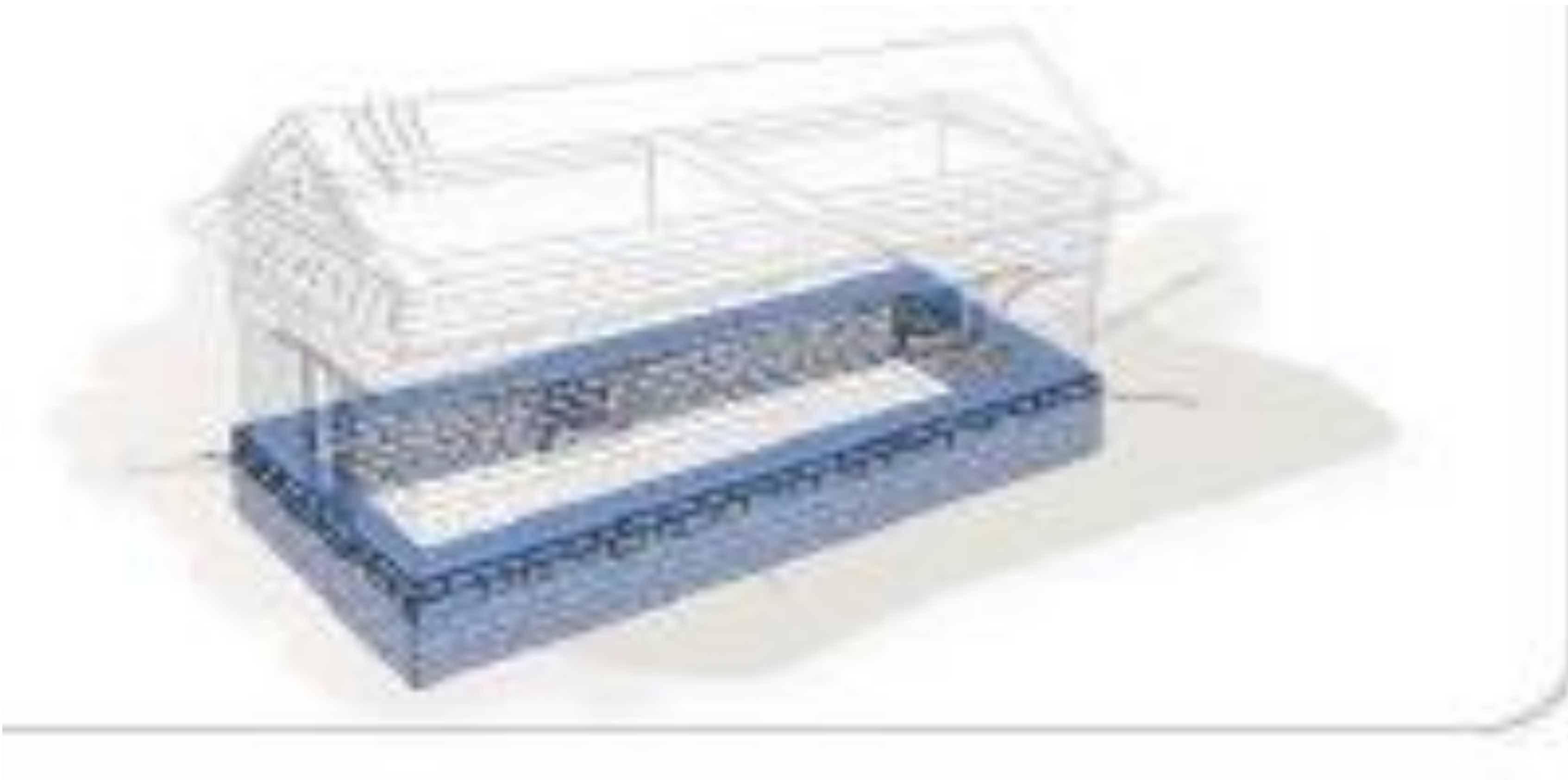
RCC Bands



10 key Message for safer Building Construction

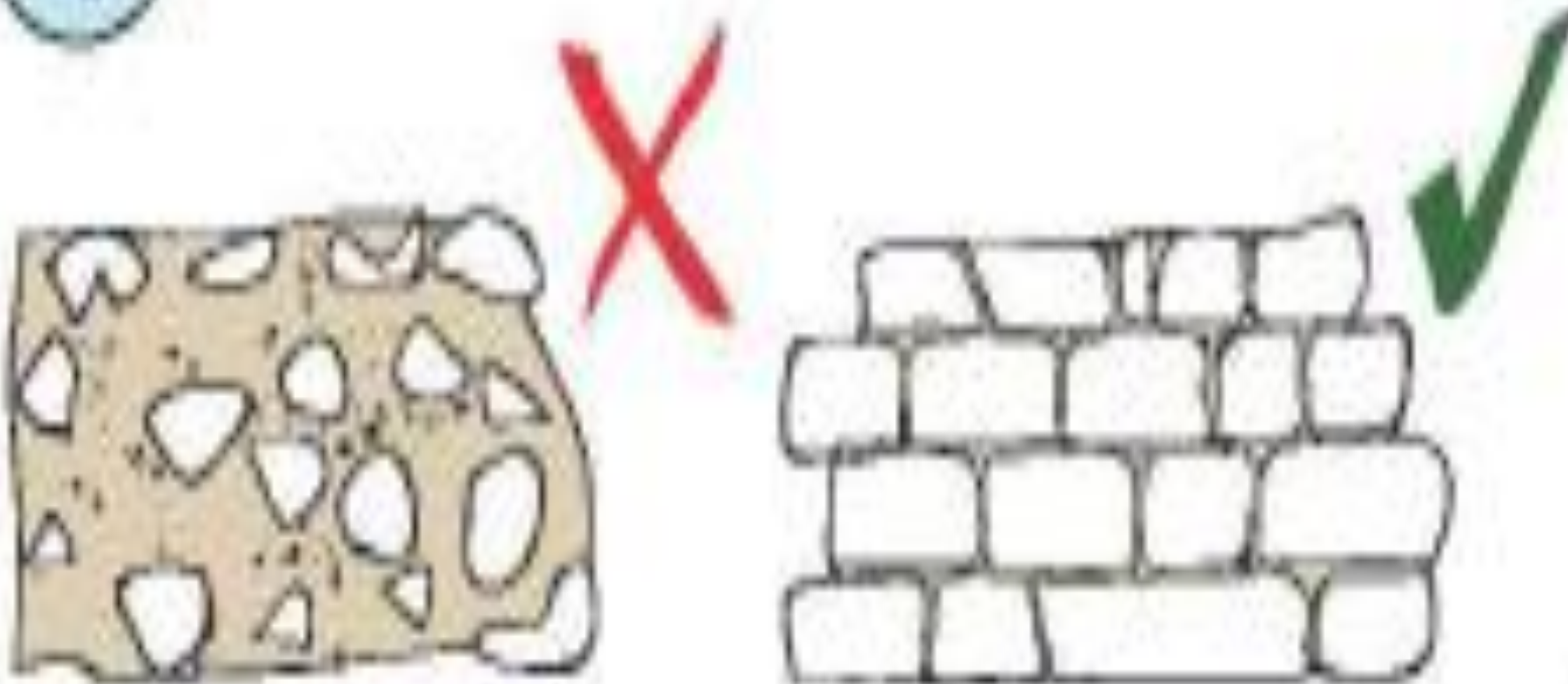
Soil invetigation

2.0 Build Strong Foundation



3.0 Through Stone

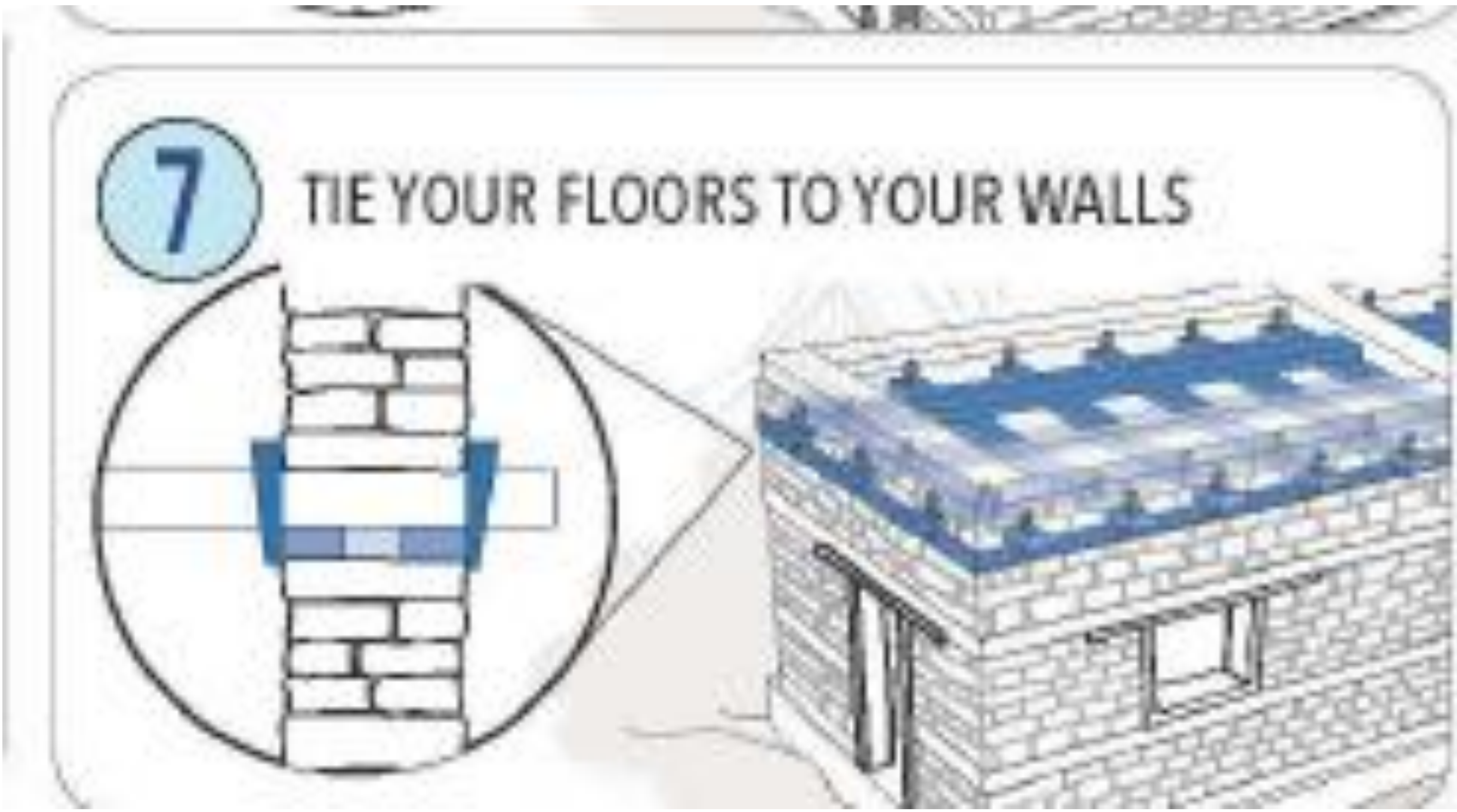
4 BUILD YOUR HOUSE WITH GOOD MATERIALS



Gable Wall

Tie the Roof with Down Part

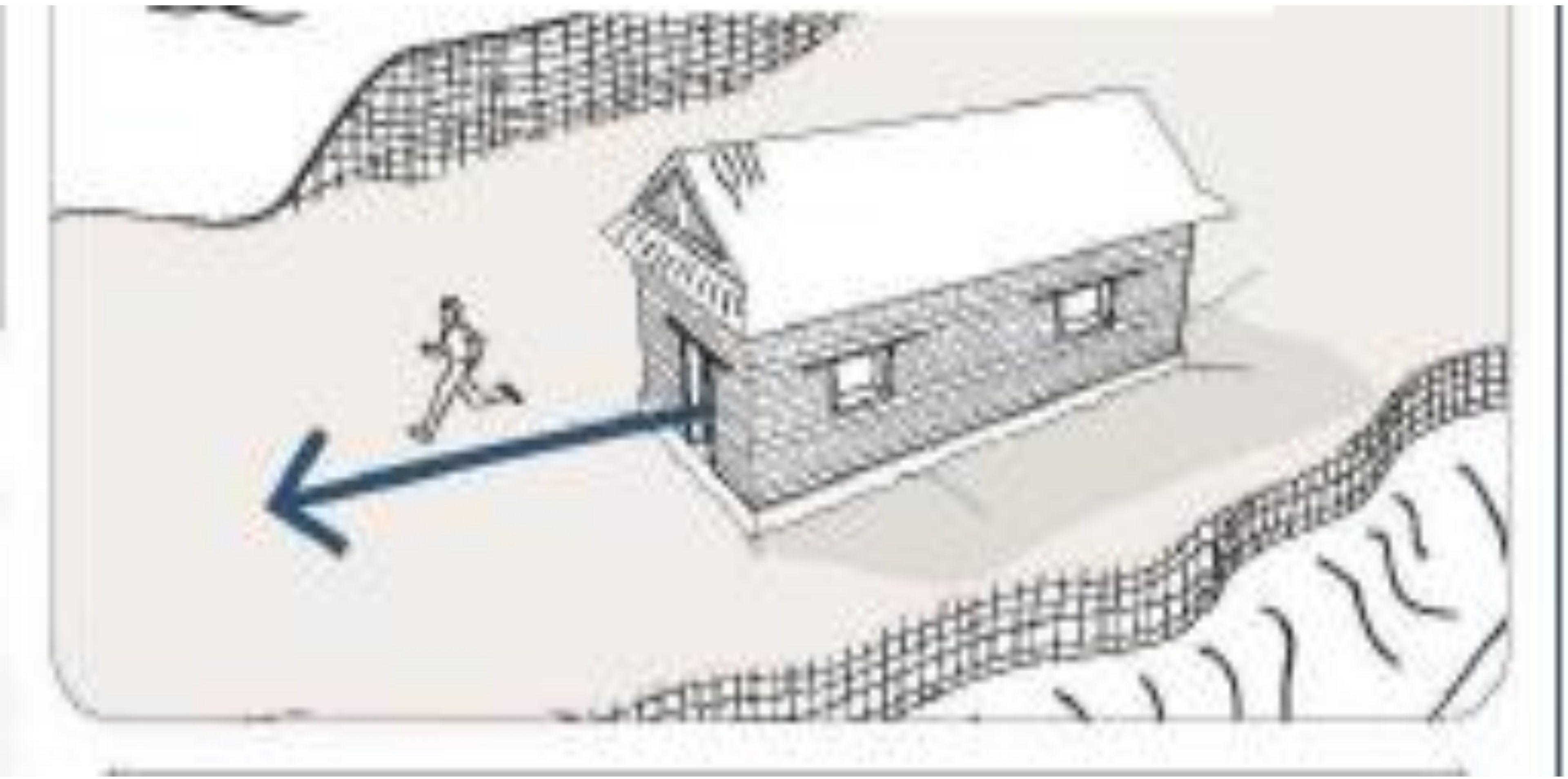
Tie floor to wall



Shape of House

9.0 Band wall Together

10.Safe site and Safe Exit



Minimum Requirements (MRs) for Stone Masonry in Cement Mortar (NBC202)				Page1
No.	Category			
1	Site Selection	A building shall not be constructed if site is:		
			✔	Geological fault or Raptured Area
			✔	Areas Susceptible to Landslide
			✔	Steep Slope > 20%
			✔	Filled Area
			✔	River Bank and Water-logged Area
2	Shape of House	No. of story	✔	Two storey+ attic, load bearing masonry buildings constructed in cement mortar
		Span of wall	✔	The span of wall shall not more than 4.5 meters
		Size of room	✔	The area of individual floor panel not more than 13.5 square metres
		Height of wall	✔	The height of wall should not be more than 3.0 meters
		Proportion	✔	The house shall be planned in square, rectangular. Avoid long and narrow structure should not be more than 3 times of its width.
3	Foundation	General	✔	The foundation trench shall be of uniform width. The foundation bed shall be on the same level throughout the foundation in flat area.
		Depth	✔	The depth of footing should not be less than 800mm for one story, 900mm for two storey.
		Width	✔	The width of footing should not be less than 600mm in medium soil condition. As depend on soil condition. Shown in detail drawings.
4	Plinth	General	✔	Provide a reinforced concrete band at plinth level, as shown in detail drawings. The top level of plinth should not be less than 300mm from existing ground level. Recommendation is 450mm.
		Height	✔	Minimum height of Plinth band is 150mm.
		Width	✔	Minimum thickness of plinth band width should be equal to wall thickness. 350mm for Stone masorny.
		Reinforcement	✔	Main reinforcement should be 4-12 dia. bars. Use 6mm diameter rings at 150mm. Hook length should be 500mm. Bars shall have a clear cover of 25mm concrete.
5	Walls	General	✔	Masonry should not be laid staggered or straggled in order to avoid continuous vertical joints. At corners or wall junctions, through vertical joints should be avoided by properly laying the masonry. It should be interlocked.
		Joints	✔	Mortar joints should not be more than 20mm and less than 10mm in thickness. The ratio recommend 1:4 (Cement: Sand).
		Through Stone	✔	Through-stone of a length equal to the full wall thickness should be used in every 600 mm lift at not more than 1.2 m apart horizontally.
		Width	✔	The minimum width of wall is 350mm for one-storey and two-storey.

6	Openings	Location	✔	Openings are to be located away from inside corners by a clear distance should not be less than 600 mm.
		Total length	✔	The total length of openings in a wall is not to exceed half of the length of the wall in single-storey construction.
		Distance	✔	The horizontal distance between two openings is to be not less than 600 mm.
		Lintel level	✔	Keep lintel level same for doors and windows.
7	Vertical Reinforcement	Location	✔	Place vertical steel bars in the wall at all corners, junctions of walls and adjacent to all doors and windows. They shall be covered with cement concrete in cavities made around them during the masonry construction.
		Reinforcement	✔	The vertical reinforcing bar for masonry is given in detail drawings. 12mm dia is minimum requirements for masonry houses.
8	Horizontal Band			Horizontal bands should be provided throughout the entire wall with minimum thickness of 75 to 150 mm at following locations:
		Sill band	✔	A continuous sill band shall be provided through all walls at the bottom level of opening (specially windows). The minimum height is 75mm.
		Lintel band	✔	A continuous lintel band shall be provided through all walls at the top level of opening. The minimum height is 150mm.
		Stitch	✔	This band shall be provided where dowel-bars are required at all corners, junctions of walls. The minimum height is 75mm.
		Roof band	✔	Roof band shall be provided at the top-level of walls, so as to integrate them properly at their ends and fix them into the walls. The minimum height is 75mm.
		Reinforcement	✔	Main reinforcement should be 4or 2-12 dia. bars. Use 6mm diameter rings at 150mm. Hook length should be 500mm. Bars shall have a clear cover of 25mm concrete.
9	Roof	Light roof	✔	Use light roof comprising wooden or steel truss covered with CGI sheets.
		Connection	✔	All members of the timber truss or joints should be properly connected as shown in detail drawings.
		Cross-tie	✔	Trusses should be properly cross-tied with wooden braces as shown in detail drawings.
		Timber	✔	Well seasoned hard wood without knots should be used for roofing, timber treatment such as use of coal tar or any other preservative can prevent timber from being decayed and attacked by insects
10	Materials	Mortar	✔	Cement sand mortar should not be leaner than 1:4 (1 part cement and 4 parts sand) for masonry and 1:6 for plaster
		Concrete	✔	The concrete mix for seismic bands should not be leaner than 1:1.5:3 (1 part cement, 1.5 parts sand and 3 parts aggregate)
		Reinforcement	✔	High Strength Deformed Bars – Fe415: High strength deformed bars with fy = 415 N/

THANK YOU !!!!!!!!!!!